Multi-Modal Human-Aware Image Caption System for Intelligent Service Robotics Applications

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***Abstract*—Image captioning is a high-level task that generates the context scenario descriptions from an image. There are many ways to implement such kinds of ability that can generate a smooth sentence about the context scenario. Because of the open and extensive domain of the dataset, seldom of them can really help people to get the meaningful information of a specified scenario. In this paper, we propose a novel framework called Human-Aware Context Generator (HACG). We adopt the concept of dividing and conquer, which combines the ability of face recognition and facial expression recognition while retaining the capability of image caption model. This model aims to offer a significant context scenario sentence to the people, especially for those who are urgent to master the overall situation.**

Keywords—image caption, face recognition, machine learning, facial expression recognition, human-aware context generator

I. Introduction

Service robot is the trend in the future. Despite the efficiency and productivity of industrial robotics, service robots show a further step towards human life. One of the most important features of service robotics is the intelligence. Among many artificial intelligence tasks, image captioning is one of the most fantastic functionalities.

Image caption [1] is a challenging mission in the domain of natural language processing and computer vision, whose purpose is to generate a description of an image automatically. It is an arduous task and not easy to generate a significant description, since the image has too much information. We need to concern a variety of element and their spatial relationship to form semantically and syntactically correct sentence.

Most studies on image caption apply the structure that Convolutional neural network (CNN) [2][3] plays the role as encoder to extract the features and long-short term memory (LSTM) [4] plays the roles as decoder to generate the sentence. each layer of the CNN extracts different feature representing the different aspect to observe the image. The other important part is LSTM, which stores every memory in the weight of the parameters and will continue to generate words of a sentence. Usually, this type of encoder-decoder structure tend to use end-to-end learning which must be trained jointly.

One of the drawbacks in end-to-end learning is that very large of dataset is needed to get the better performance. Sometimes the dataset of a specified domain maybe not easy to obtain. That will lead to a bad training result. Some people tend to make their own dataset which satisfies the scenario and knowledge they want the model to learn. The normal way is to fine-tune an image captioning model trained by the famous dataset such as MSCOCO [5], Flicker-8K [6]. Unfortunately, it is usually hard to make the neural network learn the specified things without hurting its original knowledge. This is a well-known problem called ”catastrophic forgetting” in the research field of incremental learning. The fine-tuned model will lose the original ability of description, and will prone to overfit the new dataset. Also, constructing a new dataset is very time-consuming.

Inspired by [7][8], which use the well-developed object detector to enhance its quality of the description, we propose a multi-modal fusion framework and call this model as Human-Aware Context Generator (HACG), which shows the captioning ability by extracting multi-model knowledge from the scene image. HACG combines three essential basic models, namely face recognition, facial expression recognition, and image captioning. We think that verifying the identity of a person is important and meaningful to the user, for instance, if you buy a patrol robot, and its duty is to keep the piece in the local area. The function of face recognition is necessary in this scenario since the patrol robot really needs to know the identity of a person to verify whether he/she is reasonable to appear in a certain place and time with the help of semantic topological mapping [9][10]. By the way, qualified security needs to notice the behavior of the person, so we maintain the ability of descripting the scene with the image captioning model. Having such an essential function we mention before, the patrol robot can report meaningful information to the monitor, the administrator will know who the person is and what he/she is doing. This function can be applied in a factory, a health village, or a company to maintain the public order.

The other fundamental function is facial expression recognition, because we can certify the vigor of a person according to his expression. The expression is really vital information to verify the mental state. In terms of industrial applications, for instance, a truck driver in mine pit with the poor state of mind is very dangerous for another around him. Emotion recognition can be used for early warning of fatigue driving or distracted driving. These emotions are usually high-risk group in a car accident. If a visual system is installed inside the vehicle to detect the driver’s emotions, the probability of causing a car accident can be reduced, or if some people in factory feel

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uncomfortable and try to send the distress signal, we should discover such of this situation as an emergency to deal with in time. Fig. 1 shows some applications for our HACG to the managers. Basically, the identity, the mental state, and the behavior are significant information for the surveillance since they can obtain the instant and meaningful message, and deal with the problem as soon as possible.

The objective of this research work is to generate a certain useful context sentence, including identity, expression, and behavior. At first, we use the traditional image captioning model with CNN-LSTM structure [11] to generate a sentence. For example, for the image in Fig. 1, the generated sentence is "A man is driving a car." The subject "A man" will be replaced by the person's identity with our algorithm, and expression will also be filled in the sentence. One of the example sentence generated by HACG may be "<identity> is <expression> and is playing a ball." Then the face recognition model will fill in the slot of <identity> and the expression recognition model will fill in the slot of <expression> (e.g., <identity>→Mr. Yeh, <expression> →disgust), resulting in final description "Mr. Yeh is disgust and is driving a car." which is a meaningful sentence that really can help the manager to control the situation. One special feature for our HACG model is that it allows for different face recognition models, expression recognition models, even for the image captioning model to be plugged in easily.

Contributions: our contributions can be summarized as follows:

- We propose Human-Aware Context Generator (HACG), a multi-modal fusion method that can really offer the meaningful information to the manager, answering the three vital questions: "Who is he?", "What is he doing?", and "How is he?"
- We Use a divide-and-conquer manner to deal with the problem. Our approach is divided into two stage. First, generating a hybrid template that contains the identity slot , expression slot, and the other description generated by the image captioning model. Second, the identity slot is explicitly associated with the result of the face recognition model, and so does the slot of expression. The method of divide and conquer is more easier to improve the performance.
- We present a multi-modal fusion method to obtain the specific information and format we want. Moreover, we do not lose the ability of description from image caption model, different from the common method of fine-tuning, which will lead to "catastrophic forgetting" and overfitting.

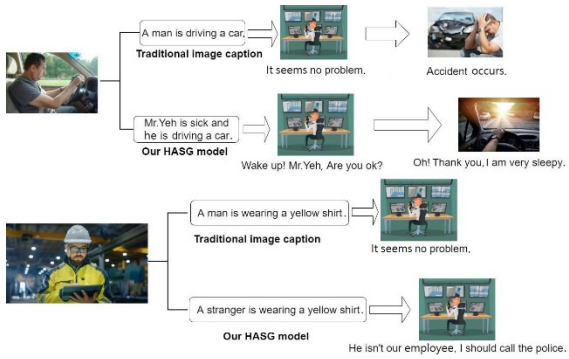

Fig. 1. Our HACG aims to help the manager to dominate the overall situation. Sometimes, a manager is busy for other things, and often ignore some details. For instance, the driver may feel uncomfortable, or the stranger is sneaking into the factory.

II. Related Work

A. Retrieval-based method

The first categories of image caption approach are the retrieval-based method [14][15], finding a similar image and use its description as the captions of the query image. The retrieval-based method can make sure the correct grammar, and not far with the correct answer. One special advantage is that the sentence is very natural, but is restricted from the limited database, the visual context in the sentence generated by using these methods sometimes is wrong. Some people improve this retrieval-based method by using the text of the retrieved image to guide the model to generate a better sentence [16]. Moreover, they also enhance their work to generate the description with more detail in object-level attention instead of in region-level attention [17].

B. Template-based method

The second categories of image caption are the template-based method [12][13], these methods use variety of detectors or classifiers to recognize the items, as well as their texture, edge, and shape in the image, and then use the fixed template to complete the whole sentence. These methods are good at detecting the key objects, and any other meaningful information you want in the image, but the sentence is lack of flexible and tend to be the rigid format.

C. Neural network-based method

The third category of image caption approach are the neural network-based method. Deep neural network becomes more and more popular in recent years. The critical point is the well-developed computer power such as GPU and the enormous of data. Famous datasets such as Flicker8k/30k, MSCOCO, and the newest dataset Conceptual Caption just was released in 2018 which consist of 3.3million image/caption pairs. The traditional model of image caption is based on CNN as an encoder and RNN as a decoder, and this structure almost uses end-to-end learning to train. Therefore, a large enough dataset will be the key to success. Inspired by the improvement of machine translation, attention mechanism is a landmark in the computer vision filed. It really improves the performance and pushes the image caption to another level. An image will be split into many regions and CNN will encode them to be many vectors instead

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of the global image as input. It helps the machine automatically attend to the relevant features when generating the next word. There are much research dedicating to the visual attention-based encoder-decoder model. For instance, Lu et al. [18] propose an adaptive attention model the will automatically determine if the next word just relies on the language model or relies on the visual signal. Generally speaking, there are a variety of attention mechanisms in recent years and constantly improve the state-of-the-art. Most common attention mechanisms are spatial attention [19][20] which make the highlight in each images region relevant to every word, and semantic attention [21][22] which use the semantic attributes from image and their captions to guide the decoder to generate the sentence, and multi-layer attention which not only use the feature extract from the last layer but also adopt the other layers. As the result of different layer corresponding to different feature map, it really get more image information and improve the attention mechanism.

Nevertheless, all the methods mentioned above have some common drawbacks. They cannot generate the sentence which is focused on the certain region in the image, and give the monitor meaningful information and specific format. We use independent face recognition model and facial expression model to extract the face feature and fill it in the template to generate the sentence more clearly and precisely than [23], which has potential error in generating the abnormal sentence by the interferences in its combined architecture. It is inconvenient to manage the overall situation and deal with any emergency. This trouble is difficult to solve only by retrieval-based method, template-based method or neural network-based method. Therefore, Our model HACG combines the neural network-based method and template-based method to conquer this problem. We use the advantage of the neural network to generate the description of the behavior, and make it flexible. On the other hand, we use the advantage of template-based to generate a specific sentence, telling a machine what information is important, and complement to the shortcoming of the neural network, which always generates the unhelpful and trivial sentence to monitor.

III. Methodology

Our fusion model algorithm aims to build a Human-Aware Context Generator (HACG). We adopt the famous encoder-decoder neural structure as an image captioning model. The encoder CNN encodes the global image to a vector, and then decoder LSTM decodes this vector to a sentence. Inspired by the machine translation using RNN-RNN structure to find the contextual clues and decode to another language sentence. Instead of using RNN as an encoder, we use CNN as an encoder, since CNN is good at processing an image and extracting the meaningful feature. Given the image is I, and the CNN-LSTM structure is trained to maximize the likelihood p(W|I), which is aimed to produce the target sentence. W={w1,w2,w3….} is the word coming from a certain dictionary and gradually assemble to a suitable sentence. We adopt the popular dataset MSCOCO as training data to update the parameter in the encoder-decoder structure neural network. Our purpose is to maximize the probability of the generating sentence, so we update the model follow the formula:

$$\theta^* = \arg\max_{\theta} \sum_{(I,S)} \log p(S|I;\theta) \tag{1}$$

Where S is the ground truth sentence, and the correct sentence is pre-marked by MSCOCO annotation, and all the sentences may be different lengths, and the I is an image, θ is the parameter in the neural network.

A. Facial Expression Recognition

The goal of facial expression recognition is to input an image, and the machine will indicate their expression status, and there are 7 kinds of expression we can classify which are respectively happy, sad, fearful, disgust, neutral, angry, and surprised. At first, we use the famous vision processing tool dlib [24]to re-process the data, in order to improve image quality, we normalize the image size and the color gradation and eliminate the noise, unify image gray value and size. The most important of all is we will first detect the face in an image and enlarge it, and cut off other unnecessary parts. These steps for data processing can build up a solid foundation for the classification.

After we improve the quality of the image, we will resize the image size to 48x48 as input through the CNN model. After the input layer, we have 128 filters and the dimensions of filters are 5x5 and 3x3, and through a serious convolution and max-pooling, and the activate function is ReLU. The final layer will extract the feature map which dimension is 384, and we flatten the vector to one dimension so that it can push into the full connected layer. The final output layer will go through the softmax algorithm and show a total of 7 results, which are normalized and the total sum is 1. We will choose the highest score as the final result and then pass it to the HACG model.

B. Face recognition

Our purpose of face recognition is to input an image. The computer will recognize the face appearing in the image and identify the person.

Since the neural network model is better than the traditional method in terms of the actual implement and performance. We adopt the famous Resnet34 model to deal with image processing. This architecture input is divided into two parts. One part is gallery images, the face in which has been registered with its corresponding name. The other part is probe images. That is the faces to be tested, and the name is not marked, so it is necessary to judge the probe image by this architecture.

By the way, the gallery images will detect the face and select the highest confident face as the encoding candidate. We will enlarge the face part and cut off other unnecessary parts. It will really enhance the accuracy of the recognition. The face will be encoded to a 128D vector. This vector represents the key features of the face. The probe images are also encoded to another 128D vector, and then we will compare them by using L2-norm judgment. That is the Euclid geometric distance of the 128D vector. The proximity judgment method use formula as

$$||A||_{\mathrm{F}} = [\textstyle\sum_i (a_i - b_i)^2]^{1/2} \tag{2}$$

A is the similarity of two vectors, a and b stands for the elements of each vector.

The overall flowchart of face recognition is shown in Fig. 2.

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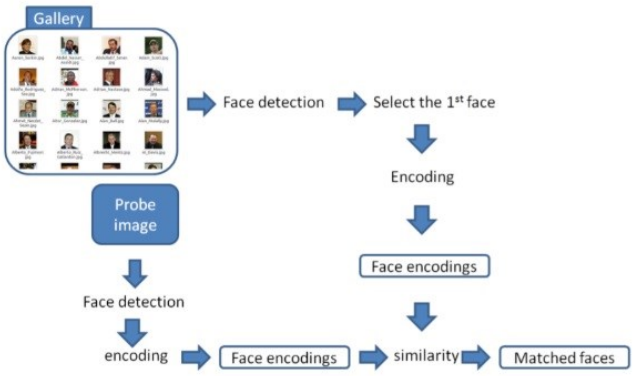

Fig. 2. The flowchart of the face recognition. If we want to register the identity, we should put the desire face images into gallery. In brief, we compare the feature of each face to recognize the identity.

C. Human-Aware Context Generator

There are two situations in the generating process. One is detecting the face of humans successfully. HACG will cut off other regions in an image, and zoom in the face part, then the face image will pass through the face recognition model and facial expression recognition model. After we obtain the information of identity and emotion, the most important step we must have to do is to combine the meaningful data adequately and generate the complete sentence syntactically. The sentence generated by HACG model will follow the template rule as "<identity> is <expression> and <behavior >(which is generated by image caption model)" . Moreover, if the detected face isn't registered by face recognition model, the identity slot will be filled in "a stranger". The format is clear and gives us useful information about the identity and mental state. The other situation is that the camera is failing to detect the face of the human. Under special circumstances, the face is difficult to capture, as a result of the lights, the quality of the camera, and so on. There are lots of things maybe obscure the face. Since there is no face image that can be processed by face recognition and facial expression model, HACG will still generate a sentence by image caption model, and straightly replace the subject of person to "a stranger". If there is indeed no person in the place. HACG will produce a sentence purely generated by image captioning model.

The flowchart of HACG is shown in Fig. 3.

IV. Experiments

In this section, we perform a comparison in the fine-tune image caption model with the traditional image caption model, and introduce the dataset made by ourselves, in order to show the commonly fine-tune method which cannot handle such kind of problem. As a result, the traditional image caption model with an opensource dataset may generate some useless sentences with confusion information.

A. Dataset

1) Training set

At the training stage, we adopt MSCOCO dataset, which contains 123287 images, and involves 40504 validation images and 82783 training images. Each image in the train set has at least 5 crowd sourced reference captions in the annotations, unlike another famous dataset Flicker8k, which contains the bounding boxes to related with natural language phrases or specific entities. Note that we build the vocabulary by the whole training set and make sure its word emerges at least five times, additionally, the identity registered by face recognition model maybe not appear in MSCOCO dataset, so we specifically add our names.

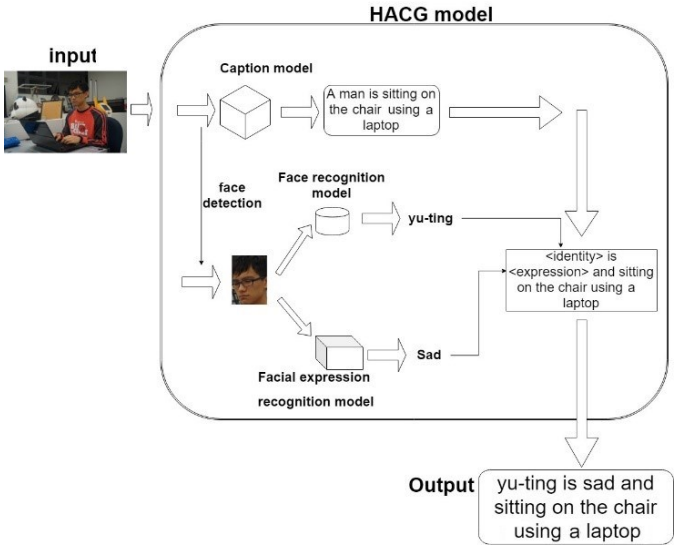

Fig. 3. The flowchart of our HACG model, we make the template and fill in the identity and expression, while the image caption model is take charge of the behavior .Each model has its contribution.

2) Fine-tuning set

The commonly method to deal with the problem that you want to obtain the significant sentence and satisfy the content you need is fine-tune. Therefore, we make the simple fine-tuning set to experiment. Total 23 images in fine-tuning set, and the reference of each image is manually constructed by the desired format."<identity> is <expression> and <behavior>" ,and there are a variety of actions, expression in each image. By the way, there is no reference of the fine-tune set which is the same as that of the test set to ensure the credibility of the experiment.

3) Test set

For the test phase, in order to avoid spending too much for labeling the reference captions for the test set. We only choose 33 images and ten of them are extracted from the MSCOCO test dataset, including/non-including humans every five sheets. If there is a person in an image, and its original description is"A man is standing on a chair", and we will judge his emotion first, for instance, he is happy, and change the label to "A stranger is happy and is standing on a chair". We haven't registered the name of a person in the MSCOCO to face recognition model, so the identity slot in MSCOCO always fill in "A stranger". If there is no person in the image, we will take the original annotations. The others are photoed and labeled by ourselves and follows the rule as "<identity> is <expression> and <behavior>". The actions and expressions of human are also variations in the test set.

B. Evaluation Metrics

How to estimate the quality of each description is challenging. Popular evaluation metrics measuring each

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description differently, such as BLEU [25], CIDEr [26], METEOR [27], ROUGE-L, and SPICE [28]. METEOR, BLEU, and ROUGE-L are famous evaluation metrics in the machine translation field, since they are originally developed to improve the translation. ROUGE-L and SPICE are purely designed for the image captioning and more associated with the consensus of a human. In order to analyze more clearly, we use all the evaluation metrics we mentioned before, which will computes similarities between descriptions and the correct answer.

C. Implementation detail

We use two kinds of machine learning library, Tensorflow, and PyTorch, since our objective is to mix the advantage of different models and get better performance. Use GPU Nvidia GeForce RTX 2080 to train, finetune, and test on HACG model, and on top of that, we resize the image to 224 x 224 as the input for image caption model, and then the face image detected by face detection is resized to 48 x48 as input for face recognition and facial expression recognition model. For the training phase, the parameter of batch size is 16, the learning rate is 0.002, and we use Adam as the optimizer. At testing phase, the HACG model needs about 2 seconds to process each given image.

D. Experimental Setup

The experience is aimed to compare the performance with the fine-tune model and the original model, and it can be divided into two parts.

1) Comparison of the performance among the HACG model, finetune model, and traditional image caption model.

2) Comparison of the performance of face recognition and facial expression recognition.

The traditional captioning model is a pure pre-trained neural image captioning model [11], and the fine-tune model is the pre-trained neural image captioning model fine-tuned by our fine-tune dataset.

E. Quantitative Evalution Results

We obey the experimental setups described before and show the score of each evaluation metric.

In table I, ”B-x” represents x-norms BLEU score, “S” represents SPICE score, “M” represents METEOR score, “C” represents CIDEr score, “R” represents ROUGE-L

TABLE I. DIFFERENT MODEL COMPARISON

Model	***B-1***	***B-2***	***B-3***	***B-4***	***S***	***M***	***C***	***R***
Our HACG	0.43	0.3	0.2	0.16	0.30	0.25	1.03	0.47
Finetune	0.34	0.21	0.14	0.10	0.11	0.13	0.89	0.32
original	0.26	0.14	0.08	0.06	0.11	0.11	0.70	0.25

The result in TABLE I. shows that the score among the finetune model, original image caption model, and our HACG model, we can apparently observe that our HACG model has better performance than others. Focus on the comparison of the finetune model and the original caption model, the finetune model scores little higher than the other since the finetune model successfully imitate the template we desire, and the original image caption model seems to generate the trivial sentence. But the content generated by finetune model tend to overfit the finetune dataset.

TABLE II. FACE RECOGNITION AND FACIAL EXPRESSION RECOGNITION

Model	Face recognition	Facial expression recognition
Our HACG	0.9412	0.353
Finetune	0.17	0.147

In TABLE II. we show the accuracy about the face recognition and facial expression recognition, we are not difficult to find that our HACG model is more robust than the finetune model.

F. Experimental result

As shown the experimental results in Table I and II, HACG model demonstrates the superior performance compared with both finetune model and original caption model. The key points are that the HACG model fuses the benefit from the face recognition model and the facial expression recognition model, then fill the results into the template to make it more easier to recognize.

Our HACG model is aimed to really reduce the burden on managers. We photo some pictures and one of the test is shown in Fig 4. For instance, someone is wearing a helmet to avoid being recognized, it is difficult to detect his face, but our HACG model will describe the dangerous action and tell the manager his identity is stranger. Another example, if the employee is playing and riding a bike in a place with lots of fragile and expensive items, it’s hard to imagine that any valuable instrument is broken, so our HACG model will remind the manager to alarm the employee. The last example in Fig. 4 is shown that if your employee always works with negative emotions, he really wants somebody can care about him or he is really uncomfortable. Our HACG model will send the message to the manager, and the manager will call somebody to take care of him. In a world, the final objective of our HACG model is to generate the meaningful sentence, which is different from the other image caption model.

Fig. 4. In order to describe more applications, we put three images as our demonstration, we want to outdo itself in any situation.

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CONCLUSION

In this paper, we propose the HACG model, which is a multi-modal fusion of the image caption, facial expression recognition, and face recognition. We combine the template-based method and neural network-based method, which provide complementary advantage/disadvantage functions and strengthen the sentence becoming more helpful and useful to the manager. The most important information to the manager are three questions ”Who is he? “, “How is he”, and “What is he doing“. So we add the identity and mental state, the most important of all we preserve the description of behavior generated by image caption model. Despite the fine-tune method is also to generate the same format and try to give meaningful information, such as identity and emotion, however, all of them tend to overfit, especially obvious for the behavior. The Fig. 5 shows the overfitting state by using the fine-tune method and compare the result of our HACG model. In the future, we maybe improve the performance of each model. This will be more easy for our HACG model since we hold the spirit of dividing and conquering.

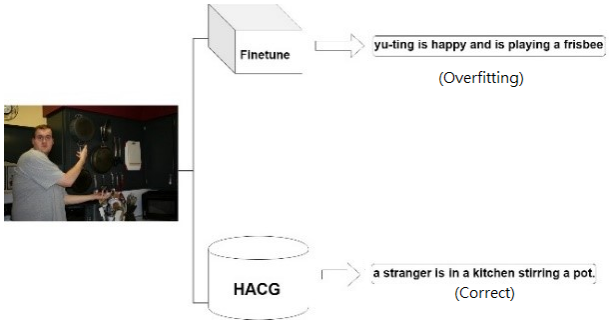

Fig. 5. We can see that the fine-tuning model obviously overfit the finetune data. Although it has the correct format, the content about the description is terrible. On the other hand, our HACG model performs robustly in the unfamiliar image.

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